# GEMINI API SETUP + PROMPT ENGINEERING GUIDE
# ApnoAI Final Project — AI/ML Track

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## PART 1: GET YOUR FREE API KEY (10 min)

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Step 1: Go to https://aistudio.google.com
Step 2: Sign in with your Google account
Step 3: Click "Get API Key" (top right corner)
Step 4: Click "Create API Key"
Step 5: Copy the key (it starts with "AIza...")
Step 6: Save it somewhere safe

IMPORTANT NOTES:
• This key is 100% FREE
• You get 15 requests per minute
• You get 1500 requests per day
• Never share this key publicly
• In Vercel, add it as an Environment Variable

TROUBLESHOOTING:
• Key not working? Wait 2 minutes after creation
• Quota exceeded? You have 1500/day — that is plenty
• Error 400? Check if you copied the full key (no spaces)
• Error 403? Enable the Gemini API in Google Cloud Console

## PART 2: ADD KEY TO YOUR CODE (5 min)

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1. Open index.html in VS Code
2. Find line ~280: const GEMINI_API_KEY = "YOUR_API_KEY_HERE";
3. Replace YOUR_API_KEY_HERE with your actual key
4. Keep the quotes around the key
5. Save (Ctrl+S)
6. Double-click index.html to test

EXAMPLE:
const GEMINI_API_KEY = "AIzaSyD1234567890abcdefghijklmnop";

## PART 3: UNDERSTAND PROMPT ENGINEERING (30 min)

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The SYSTEM_PROMPT variable (around line 290) is the BRAIN of your AI.
It has 4 parts — this is the Prompt Engineering Framework:

1. ROLE — Who is the AI?
"You are an expert career coach and ATS specialist..."

→ Gives the AI a persona and expertise level
→ Try: "You are a tech recruiter at Google with 10 years experience"

2. TASK — What should it do?
"Analyze the provided resume against the job description..."

→ Clear, specific instruction
→ Try: "Compare the resume skills with job requirements and score 0-100"

3. OUTPUT FORMAT — How should it respond?
"Return your analysis as a JSON object with these exact fields..."

→ Forces structured, parseable output
→ Try: "Return as a table with columns: Skill, Present, Missing"

4. RULES — What are the constraints?
"Be SPECIFIC and DATA-BACKED. Use numbers where possible..."

→ Prevents generic, useless responses
→ Try: "Keep each item under 20 words. No fluff."

PART 4: CUSTOMIZE YOUR PROMPT (20 min)
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Try these prompt variations and see how outputs change:

OPTION A: Tech Industry Specialist
"You are a senior software engineer at a FAANG company reviewing resumes for a mid-level developer role. Focus on: coding skills, system design, LeetCode experience, open source contributions. Be strict but fair."

OPTION B: Fresh Graduate Advisor
"You are a campus recruiter who has hired 500+ fresh graduates. Focus on: internships, projects, CGPA, extracurriculars. Be encouraging but honest about gaps."

OPTION C: Career Changer Guide
"You are a career transition coach specializing in helping professionals switch industries. Focus on: transferable skills, relevant experience, learning agility. Be supportive and highlight hidden strengths."

TEST EACH: Paste the same resume + job description and compare outputs. Notice how the ROLE changes the tone and focus of feedback.

PART 5: CUSTOMIZE THE UI (30 min)
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Find the :root section at the top of the file (around line 15):

```
:root {
  --primary: #0f172a;      ← Change header color
  --accent: #ea580c;       ← Change button/highlight color
  --bg: #f8fafc;           ← Change page background
}
```

COLOR IDEAS FOR DIFFERENT BRANDS:
• Tech/Startup: --primary: #1e293b; --accent: #3b82f6;
• Creative: --primary: #7c3aed; --accent: #ec4899;
• Professional: --primary: #0f172a; --accent: #10b981;
• Finance: --primary: #1e3a5f; --accent: #d4af37;

Also customize:
• Header title (line ~230): Change "AI Career Copilot" to your brand
• Footer (line ~310): Change "[Your Name]" to your actual name
• Tips box text (line ~270): Write your own pro tip

PART 6: DEPLOY ON VERCEL (15 min)
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1. Go to github.com → Sign in
2. Click "+" → "New repository"
3. Name: ai-career-copilot-yourname (small letters, hyphens)
4. Description: "AI Career Copilot — Resume Analyzer using Gemini API"
5. Public → Do NOT check "Add README" → Create repository
6. Click "uploading an existing file" → drag index.html → Commit
7. Go to vercel.com → Sign in with GitHub
8. Click "Add New" → "Project"
9. Find your repo → Import
10. Framework Preset: "Other" → Deploy
11. Wait 30-60 seconds for confetti
12. Copy your live URL

PART 7: HIDE YOUR API KEY (10 min) — IMPORTANT!
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Your API key is currently visible in the code. Fix this:

1. Vercel dashboard → your project → Settings → Environment Variables

2. Add new variable:
   Name = GEMINI_API_KEY
   Value = your actual key
3. In index.html, change:
   const GEMINI_API_KEY = process.env.GEMINI_API_KEY;

   NOTE: process.env only works in server-side code. For static HTML,
   use a simple obfuscation or accept the risk for a student project.

4. Redeploy on Vercel

For student projects, keeping the key in code is acceptable since:
• It's a free tier key with low limits
• It's a learning project, not production
• You can regenerate the key anytime

PART 8: ADVANCED FEATURES (+10 XP)
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Choose ONE to add for bonus XP:

A. PDF UPLOAD
  • Use pdf.js library: https://mozilla.github.io/pdf.js/
  • Extract text from uploaded PDF resumes
  • Ask ChatGPT: "How to add PDF upload with pdf.js to HTML"

B. SAVE HISTORY
  • Use localStorage to save past analyses
  • Show history in a sidebar
  • Ask Claude: "Give me localStorage code to save analysis history"

C. MULTI-LANGUAGE
  • Add Hindi/English toggle
  • Translate prompts using Gemini
  • Ask ChatGPT: "How to add language toggle with Gemini API"

D. EXPORT AS PDF
  • Use jsPDF library
  • Download results as PDF
  • Ask ChatGPT: "How to export HTML results to PDF with jsPDF"

E. DARK MODE
  • Add toggle button
  • Change CSS variables on click
  • Reuse Task 2 dark mode logic

PART 9: TROUBLESHOOTING
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PROBLEM: "No response from AI"
FIX: Check API key is correct, wait 1 minute (rate limit), try again

PROBLEM: "JSON parse error"
FIX: AI returned invalid JSON. Improve your prompt's format rules.
Add: "Return ONLY the JSON object, no other text."

PROBLEM: "Generic/boring responses"
FIX: Your prompt is too vague. Add specific rules like:
"Use numbers. Give examples. Be critical. No fluff."

PROBLEM: "Slow response"
FIX: Gemini free tier has rate limits. Wait 30 seconds between requests.

PROBLEM: "Deployment failed on Vercel"
FIX: File must be named exactly index.html (lowercase).

PART 10: SUBMISSION CHECKLIST
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Before submitting, verify:
API key is working (test with a real resume)
☐ All 5 result sections display correctly
☐ You customized at least the business name and colors
☐ You tested with 3 different resumes
☐ Live URL works in incognito mode
☐ GitHub repo has your code
☐ You wrote 1 line about AI usage in submission

NEED HELP?
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• Post your Vercel URL in WhatsApp group for peer feedback
• Mentors review within 48 hours
• For personal issues: hello.apnoai@gmail.com

GOOD LUCK! 🚀
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